Devin Wong

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Education

University of California, San Diego **Expected June 2028**
Bachelor of Science, Mechanical Engineering *La Jolla, CA*
Cumulative GPA: **3.71** / 4.00
Available for an internship from **May 2027** to **September 2027**

Experience

Tesla **Sep 2026 – Jun 2027**
Tooling & Process Engineering Intern - Castings *Lathrop, CA*

- Building a shot-sleeve tolerance standard across **8** die programs using worst-case and nominal stack-ups
- Redesigning trim-die coupon collection in **CATIA V5** to eliminate operator cell entry from concept through installation
- Leading a shot-sleeve liner-life study, benchmarking a sister plant to identify root causes of early replacement
- Evaluating shot-tip cooling geometry and material section to extend tip replacement intervals
- Running a trim-cutter coating trade study against current lubricant practice to reduce aluminum buildup

Roche **Jun 2026 – Sep 2026**
Mechanical Engineering Intern - Sensor Module Development *Santa Clara, CA*

- Owned a trade study of **four** sensor-module geometries, selecting the one that cut shear **70%** and pressure drop **72%**
- Built grid-independent CFD in **SolidWorks Flow Simulation** (**15M** cells), validated against a **PD-pump** rig
- Ran steady-state thermal analysis across sensor interfaces, improving temperature uniformity by a **factor of two**
- Designed and prototyped a confocal microscope fixture for precise z-position control within a **30 mm** envelope

Triton Racing (Formula SAE) - University of California, San Diego **Oct 2025 – Jun 2026**
Mechanical Design & Analysis Engineer - Aerodynamics *San Diego, CA*

- Owned the TR26 rear-wing structures package end-to-end, setting load cases and driving material/layup trade studies
- Performed structural FEA in **ANSYS Mechanical**, validating parts under **200+ lb** loads to a target **FOS ∼1.5**
- Conducted 2D steady-state CFD in **ANSYS Fluent** to optimize pressure distributions, slot gaps, and element sizing
- Generated CAM toolpaths in **Fusion 360** to machine high-density foam molds on a 3-axis CNC router
- Authored **10+** detailed **GD&T** drawings (ASME Y14.5-2018), improving fabrication and alignment accuracy

Projects

Triton Racing | Front Wing 2D CFD | *ANSYS Fluent, Mesh Convergence, Parametric Optimization* **Mar 2026**

- Ran a two-element airfoil study (**NACA 4412 + S1223**) in **ANSYS Fluent** (k-ω SST) to **Cl 2.74**, **L/D 11.0**
- Established mesh independence via a four-level study and Richardson-extrapolation **GCI of 2.4%**
- Drove design with a **50-point** sweep across **5** parameters, identifying element spacing as the dominant sensitivity

Triton Racing | Rear Wing Spar | *SolidWorks, ANSYS ACP, Fusion 360, CNC Router, Wet Layup* **Jan 2026**

- Designed a **31"** carbon-fiber spar carrying a **400 N/m** distributed predicted design load, validated to a FOS of **5.5**
- Created CAM toolpaths to CNC-machine a foam mold to **0.004**" stepdown tolerance, enabling clean release post layup
- Conducted a **4-ply** bidirectional carbon-fiber layup holding peak deflection to **0.38 mm** under predicted load
- Validated laminate stiffness against ANSYS Composite Analysis with a physical **3-point bend test**

Triton Racing | Rear Wing Mounts | *SolidWorks, ANSYS Mechanical, FEA, Bolt Theory, Water Jet* **Nov 2025**

- Reduced **6061-T6** rear-wing mount mass by **77%** through topology optimization while meeting a **200 N** load
- Validated FEA against hand calculations for bending stress and deflection under design loads
- Resolved boundary-condition singularities and achieved <**0.5%** mesh convergence across five refinements (**2.2M cells**)
- Designed chassis-interface bolted joints for shear and bearing loads; waterjet-cut final mounts

Technical Skills

Software: SolidWorks (Flow Simulation, FEA), CATIA V5/3DX, ANSYS Mechanical/ACP/Fluent, Fusion 360
Design & Analysis: GD&T, Tolerance Stack-Ups, DFM/DFA, Parametric Modeling, FEA, CFD, MATLAB
Machining: CNC Router, Haas VF0, Laser Cutter, 3D Printer, Water Jet, Hot-Wire Cutter